

Logic and Algorithms -- Guided Notes ANSWER KEY

Unit tp-3.8 | CompTIA Network+ N10-009 Obj. FC0-U71 3.8 | N10-008 FC0-U71 3.8

For instructor use / distribute after students complete the notes.

Question 1

An _____ is a step-by-step set of instructions for solving a problem or completing a task. Every computer program is built on _____. Key properties of a good algorithm are that it must be _____ (has a definite end), produces the _____ output for the same input, and is _____.

Answer: Algorithm; algorithms; finite; correct (same); unambiguous.

Question 2

The three fundamental control structures in programming are: _____ (executes instructions one after another), _____ (chooses between paths based on a condition), and _____ (repeats a block of code while or until a condition is met).

Answer: Sequence; selection (if/else); iteration (loops).

Question 3

A Boolean expression evaluates to either _____ or _____. Boolean operators include _____ (both conditions must be true), _____ (at least one condition must be true), and _____ (reverses the truth value). These are the building blocks of all _____ statements in code.

Answer: True; false; AND; OR; NOT; conditional (if/else).

Question 4

A _____ loop runs code a fixed number of times or over each item in a list. A _____ loop runs as long as a condition is true -- if the condition never becomes false, a _____ loop occurs. A _____ loop always runs at least once before checking the condition.

Answer: For; while; infinite; do-while.

Question 5

A _____ is a named block of reusable code that performs a specific task. Functions accept _____ as input values and may _____ an output value. Using functions reduces _____ (writing the same code repeatedly) and makes programs easier to test and _____.

Answer: Function (subroutine/procedure); parameters; return; redundancy (duplication); maintain.

Question 6

Pseudocode is an _____ description of an algorithm using plain language rather than a specific programming language. A _____ chart visually represents the flow of an algorithm using standard symbols: rectangles for _____, diamonds for _____, and parallelograms for input/output.

Answer: Informal; flowchart; processes (actions); decisions.

Question 7

A sorting algorithm arranges a list of items in a specific _____ (ascending or descending). Bubble sort repeatedly _____ adjacent items if they are in the wrong order -- it is simple but _____ for large data sets. Quick sort and merge sort are far more _____ for large datasets.

Answer: Order; swaps (compares); slow (inefficient); efficient.

Question 8 -- Real World Application

REAL WORLD: A student is writing a program that reads a user's grade (0-100) and prints a letter grade (A, B, C, D, or F). Write the logic in pseudocode using if/else selection statements to determine the correct letter grade. Use standard letter grade ranges (A = 90-100, B = 80-89, C = 70-79, D = 60-69, F = 0-59).

Answer: INPUT grade IF grade >= 90 THEN PRINT 'A' ELSE IF grade >= 80 THEN PRINT 'B' ELSE IF grade >= 70 THEN PRINT 'C' ELSE IF grade >= 60 THEN PRINT 'D' ELSE PRINT 'F' END IF Key points: conditions are checked from highest to lowest so each threshold only needs a single comparison. If checked lowest-first, every grade >=60 would hit the D branch before reaching A/B/C.